

```
In [8]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [12]: df = pd.read_csv("train.csv")

dfhead()
```

```
Out[12]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

```
In [17]: import pandas as pd

train = pd.read_csv("train.csv")
gs = pd.read_csv("gender_submission.csv")
test = pd.read_csv("test.csv")

# Merge datasets on PassengerId
merged = train.merge(gs, on="PassengerId", how="left") \
        .merge(test, on="PassengerId", how="left")

# Save to CSV
merged.to_csv("final_dataset.csv", index=False)

print("Merged dataset created successfully!")
print(merged.head())
```

Merged dataset created successfully!

	PassengerId	Survived_x	Pclass_x	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name_x	Sex_x	Age_x	SibSp_x	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch_x	Ticket_x	Fare_x	...	Pclass_y	Name_y	Sex_y	Age_y	\
0	0	A/5 21171	7.2500	...	NaN	NaN	NaN	NaN	
1	0	PC 17599	71.2833	...	NaN	NaN	NaN	NaN	
2	0	STON/O2. 3101282	7.9250	...	NaN	NaN	NaN	NaN	
3	0	113803	53.1000	...	NaN	NaN	NaN	NaN	
4	0	373450	8.0500	...	NaN	NaN	NaN	NaN	

	SibSp_y	Parch_y	Ticket_y	Fare_y	Cabin_y	Embarked_y
0	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN

[5 rows x 23 columns]

```
In [23]: import pandas as pd

train = pd.read_csv("train.csv")
gs = pd.read_csv("gender_submission.csv")
test = pd.read_csv("test.csv")

# Store merged data in df
df = train.merge(gs, on="PassengerId", how="left") \
        .merge(test, on="PassengerId", how="left")

# Display first 5 rows
print(df.head())

# Display entire DataFrame
print(df)
```

	PassengerId	Survived_x	Pclass_x	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name_x	Sex_x	Age_x	SibSp_x	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch_x	Ticket_x	Fare_x	...	Pclass_y	Name_y	Sex_y	Age_y	\
0	0	A/5 21171	7.2500	...	NaN	NaN	NaN	NaN	
1	0	PC 17599	71.2833	...	NaN	NaN	NaN	NaN	
2	0	STON/O2. 3101282	7.9250	...	NaN	NaN	NaN	NaN	
3	0	113803	53.1000	...	NaN	NaN	NaN	NaN	
4	0	373450	8.0500	...	NaN	NaN	NaN	NaN	

	SibSp_y	Parch_y	Ticket_y	Fare_y	Cabin_y	Embarked_y
0	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN

[5 rows x 23 columns]

	PassengerId	Survived_x	Pclass_x	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

	Name_x	Sex_x	Age_x	\
0	Braund, Mr. Owen Harris	male	22.0	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	
2	Heikkinen, Miss. Laina	female	26.0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	
4	Allen, Mr. William Henry	male	35.0	
..	...	...	...	
886	Montvila, Rev. Juozas	male	27.0	
887	Graham, Miss. Margaret Edith	female	19.0	
888	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	
889	Behr, Mr. Karl Howell	male	26.0	
890	Dooley, Mr. Patrick	male	32.0	

SibSp_x	Parch_x	Ticket_x	Fare_x	...	Pclass_y	Name_y	Sex_y	\
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0	1	0	A/5	21171	7.2500	...	NaN	NaN	NaN
1	1	0	PC	17599	71.2833	...	NaN	NaN	NaN
2	0	0	STON/O2.	3101282	7.9250	...	NaN	NaN	NaN
3	1	0		113803	53.1000	...	NaN	NaN	NaN
4	0	0		373450	8.0500	...	NaN	NaN	NaN
..	...	...		...	...	...	...	...	...
886	0	0		211536	13.0000	...	NaN	NaN	NaN
887	0	0		112053	30.0000	...	NaN	NaN	NaN
888	1	2	W./C.	6607	23.4500	...	NaN	NaN	NaN
889	0	0		111369	30.0000	...	NaN	NaN	NaN
890	0	0		370376	7.7500	...	NaN	NaN	NaN

	Age_y	SibSp_y	Parch_y	Ticket_y	Fare_y	Cabin_y	Embarked_y
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN
..	...	...	...	...	...	...	...
886	NaN	NaN	NaN	NaN	NaN	NaN	NaN
887	NaN	NaN	NaN	NaN	NaN	NaN	NaN
888	NaN	NaN	NaN	NaN	NaN	NaN	NaN
889	NaN	NaN	NaN	NaN	NaN	NaN	NaN
890	NaN	NaN	NaN	NaN	NaN	NaN	NaN

[891 rows x 23 columns]

```
In [27]: df.to_csv("final_dataset.csv", index=False)
```

```
In [28]: df = pd.read_csv("final_dataset.csv")
```

```
In [29]: df.head()
```

Out[29]:

	PassengerId	Survived_x	Pclass_x	Name_x	Sex_x	Age_x	SibSp_x	Parch_x	Ticket_
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/2117
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 1759
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/Oz 310128
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	11380
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	37345

5 rows × 23 columns



In [30]:

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 23 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived_x      891 non-null    int64
2   Pclass_x        891 non-null    int64
3   Name_x          891 non-null    object
4   Sex_x           891 non-null    object
5   Age_x           714 non-null    float64
6   SibSp_x         891 non-null    int64
7   Parch_x         891 non-null    int64
8   Ticket_x        891 non-null    object
9   Fare_x          891 non-null    float64
10  Cabin_x         204 non-null    object
11  Embarked_x      889 non-null    object
12  Survived_y      0 non-null      float64
13  Pclass_y        0 non-null      float64
14  Name_y          0 non-null      float64
15  Sex_y           0 non-null      float64
16  Age_y           0 non-null      float64
17  SibSp_y         0 non-null      float64
18  Parch_y         0 non-null      float64
19  Ticket_y        0 non-null      float64
20  Fare_y          0 non-null      float64
21  Cabin_y         0 non-null      float64
22  Embarked_y      0 non-null      float64
dtypes: float64(13), int64(5), object(5)
memory usage: 160.2+ KB
```

```
In [31]: df.describe()
```

Out[31]:

	PassengerId	Survived_x	Pclass_x	Age_x	SibSp_x	Parch_x	Fare_x
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204200
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [37]: df['Survived_x'].value_counts()
```

```
Out[37]: Survived_x
0      549
1      342
Name: count, dtype: int64
```

```
In [41]: df['Sex_x'].value_counts()
```

```
Out[41]: Sex_x
male      577
female    314
Name: count, dtype: int64
```

```
In [52]: print(df.isnull().sum())
```

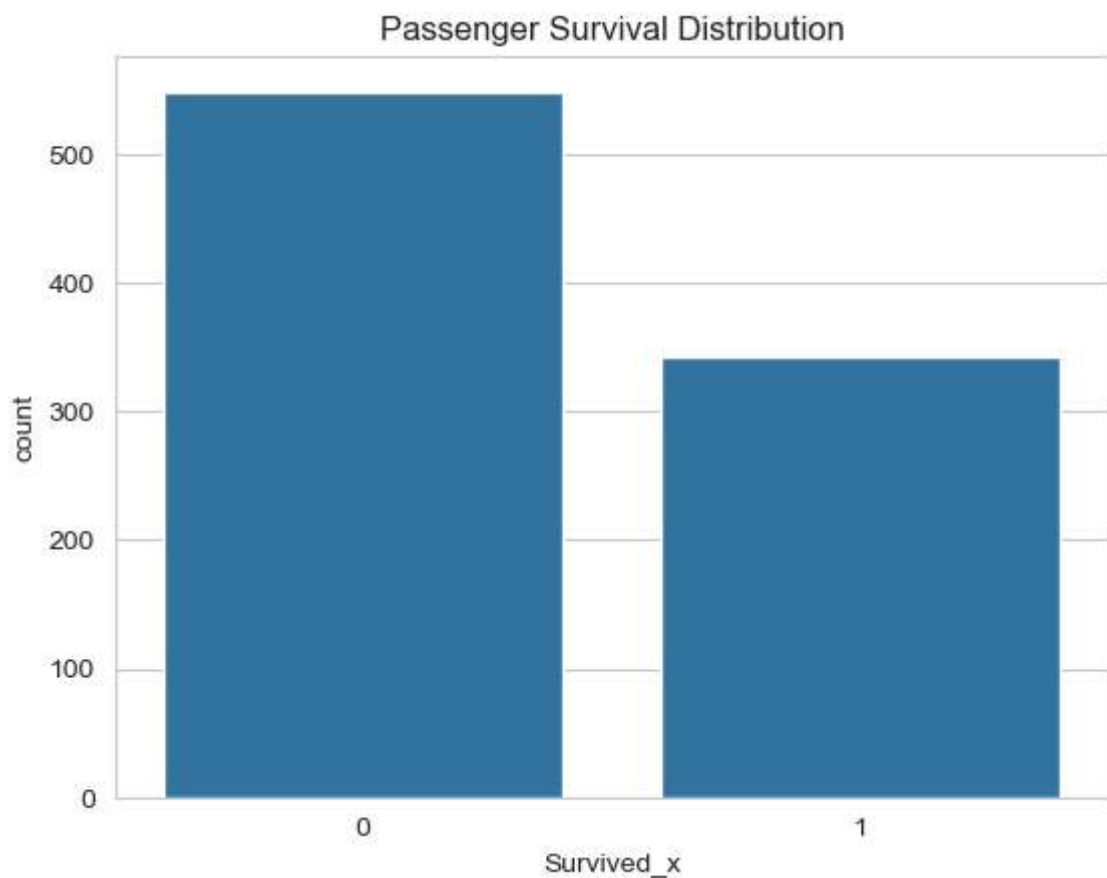
```
PassengerId      0
Survived_x        0
Pclass_x          0
Name_x            0
Sex_x             0
Age_x             0
SibSp_x           0
Parch_x           0
Ticket_x          0
Fare_x            0
Embarked_x        0
Survived_y      891
Pclass_y        891
Name_y          891
Sex_y           891
Age_y           891
SibSp_y         891
Parch_y         891
Ticket_y        891
Fare_y          891
Embarked_y      891
dtype: int64
```

```
In [72]: df.drop([
    'Survived_y',
    'Pclass_y',
    'Name_y',
    'Sex_y',
    'Age_y',
    'SibSp_y',
    'Parch_y',
    'Ticket_y',
    'Fare_y',
    'Embarked_y'
], axis=1, inplace=True)
```

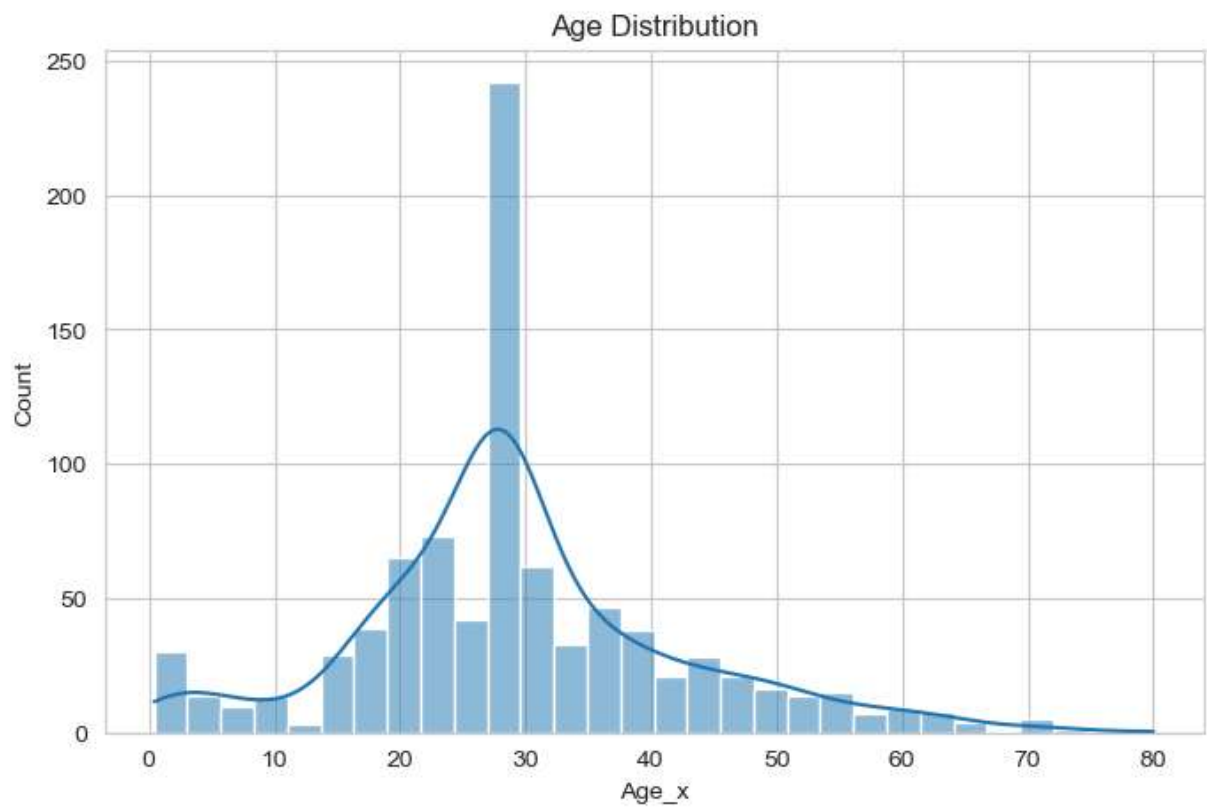
```
In [73]: print(df.isnull().sum())
```

```
PassengerId    0
Survived_x      0
Pclass_x        0
Name_x          0
Sex_x           0
Age_x           0
SibSp_x         0
Parch_x         0
Ticket_x        0
Fare_x          0
Embarked_x      0
dtype: int64
```

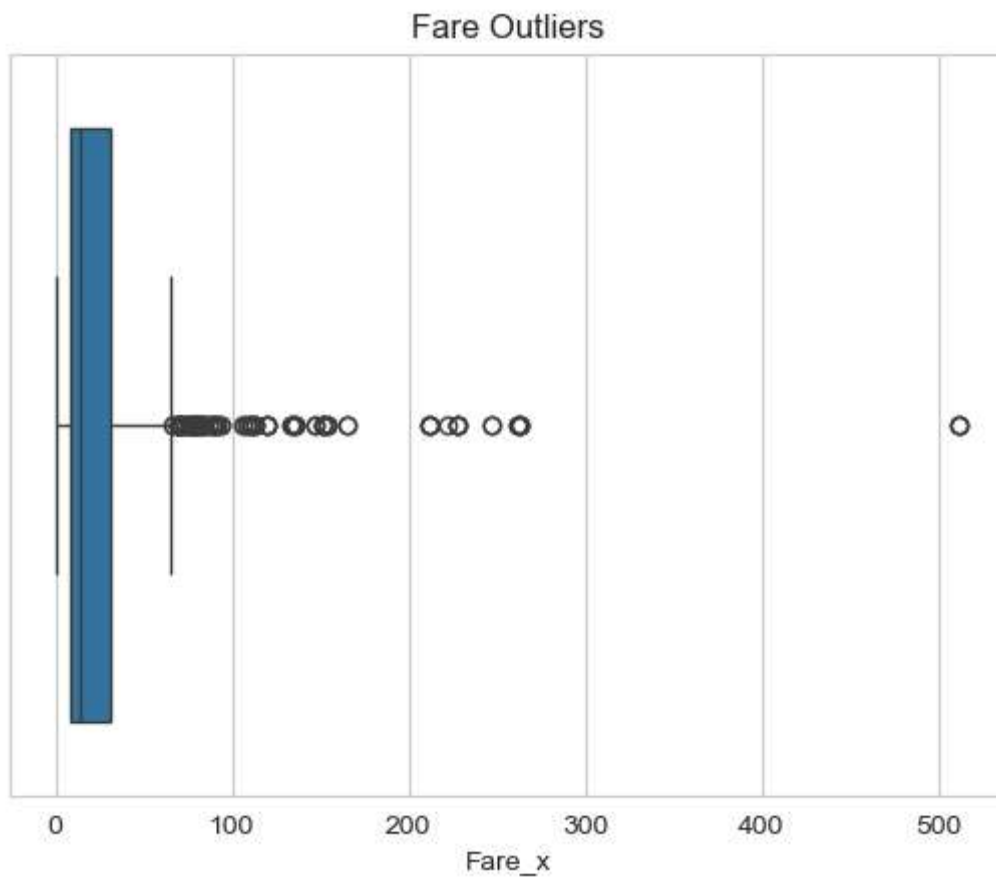
```
In [55]: sns.countplot(x='Survived_x', data=df)
plt.title("Passenger Survival Distribution")
plt.show()
```



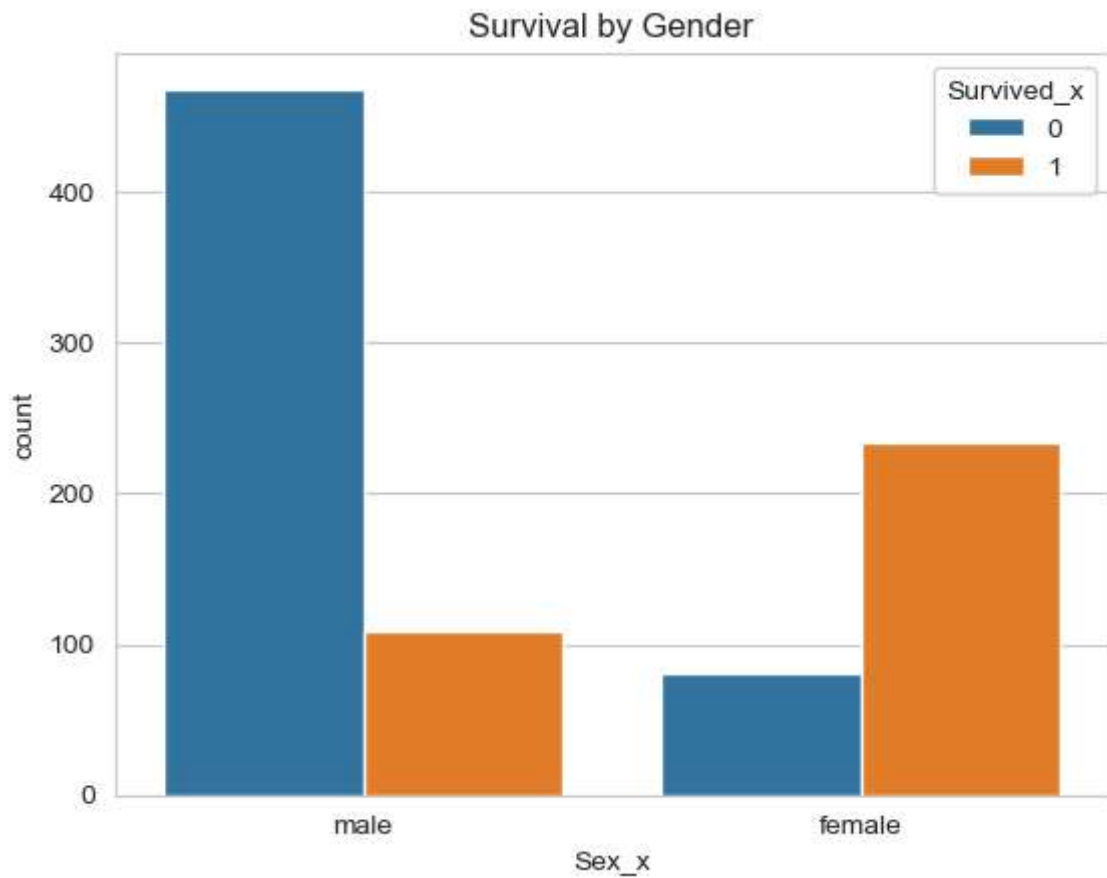
```
In [57]: plt.figure(figsize=(8,5))
sns.histplot(df['Age_x'], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()
```



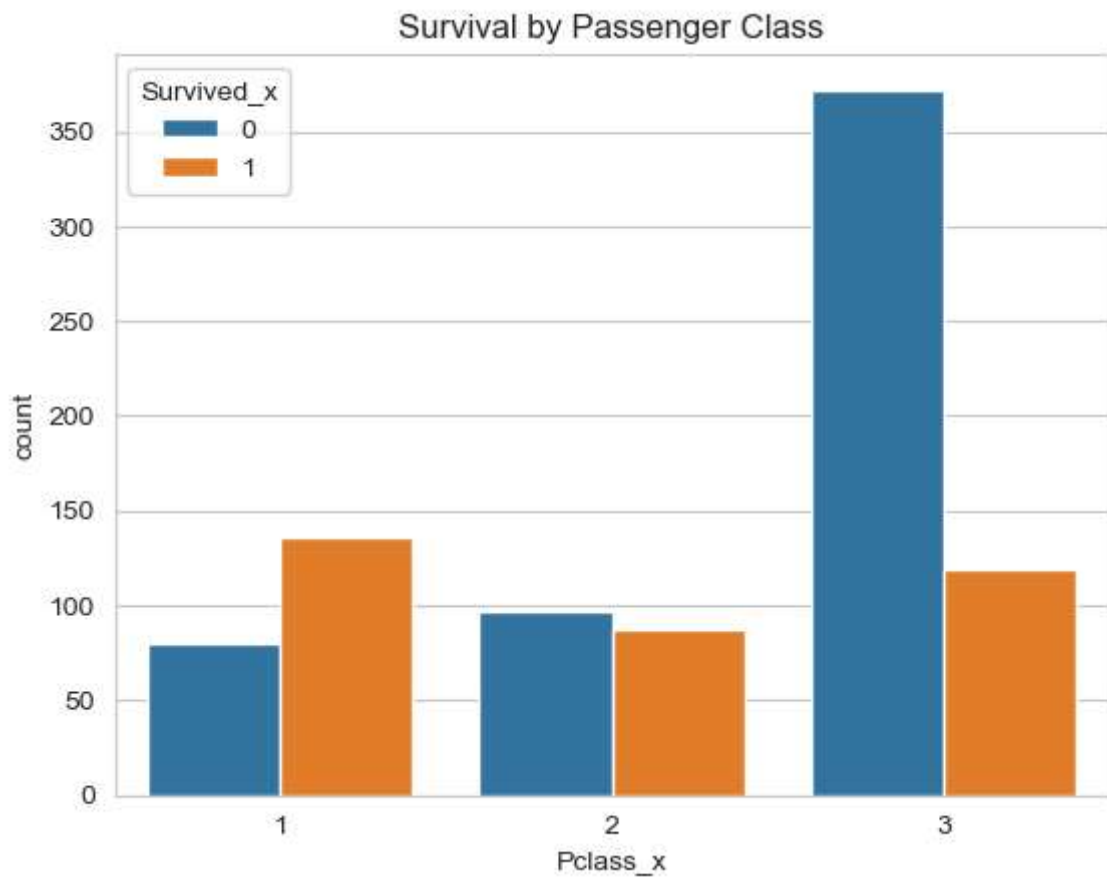
```
In [61]: sns.boxplot(x=df['Fare_x'])  
plt.title("Fare Outliers")  
plt.show()
```



```
In [62]: sns.countplot(x='Sex_x', hue='Survived_x', data=df)
plt.title("Survival by Gender")
plt.show()
```



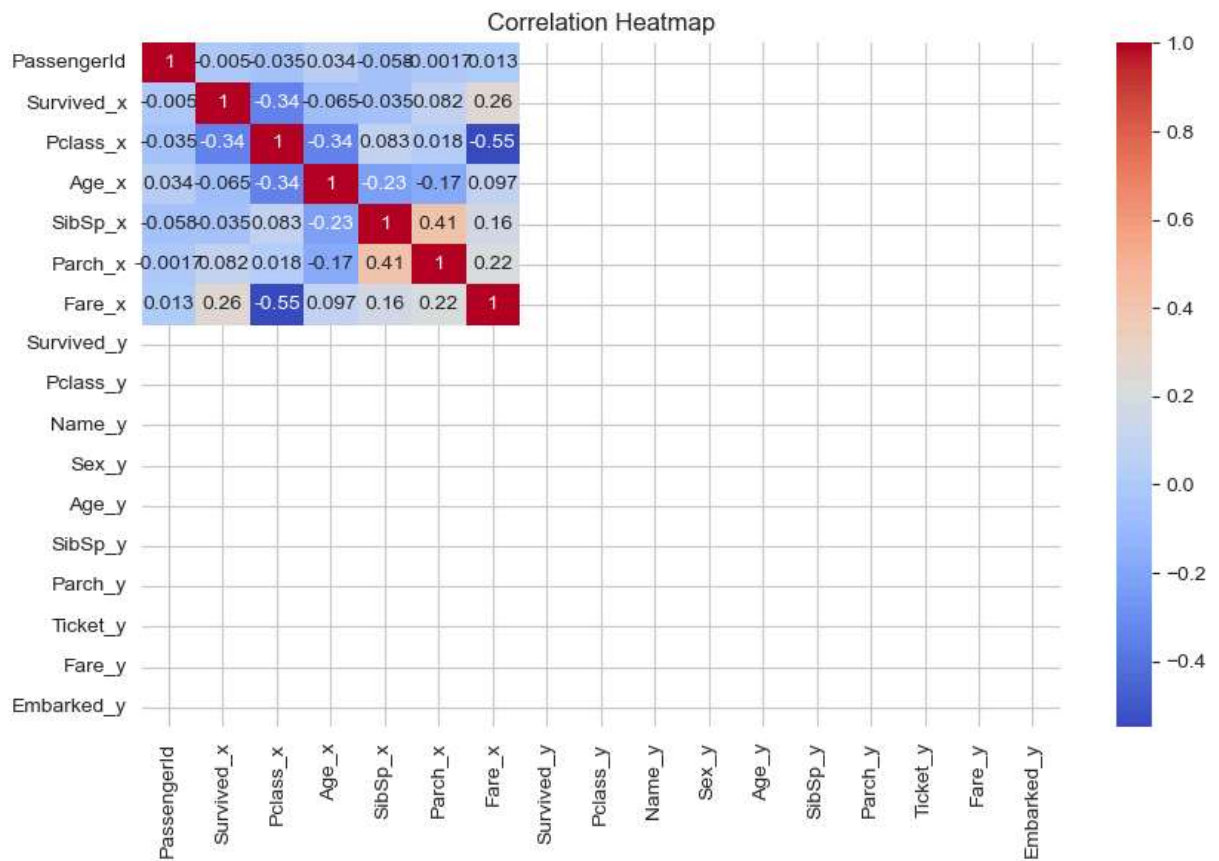
```
In [63]: sns.countplot(x='Pclass_x', hue='Survived_x', data=df)
plt.title("Survival by Passenger Class")
plt.show()
```



```
In [65]: numeric_df = df.select_dtypes(include=['number'])
```

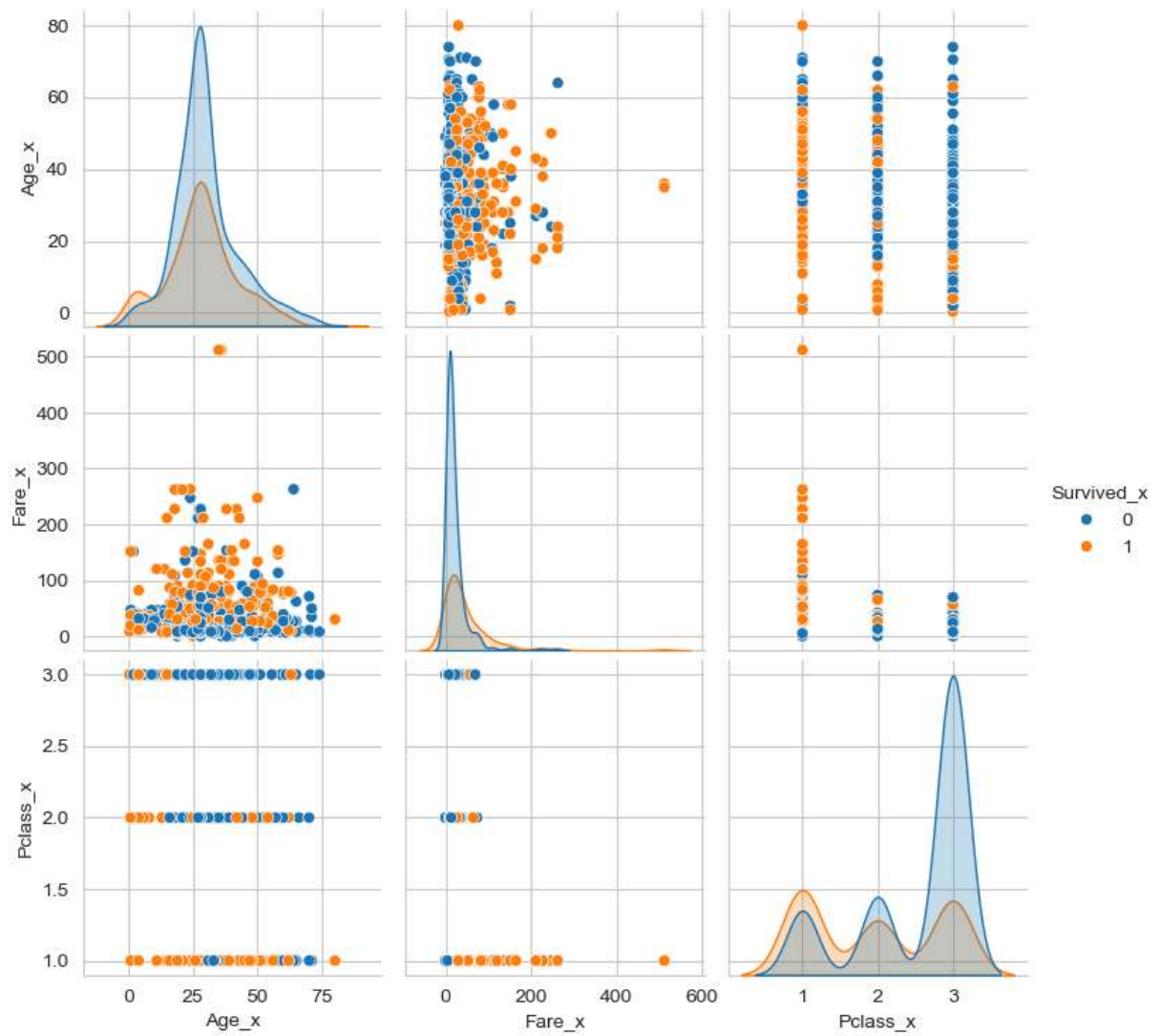
```
plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap='coolwarm')

plt.title("Correlation Heatmap")
plt.show()
```



```
In [70]: sns.pairplot(
          df[['Age_x', 'Fare_x', 'Pclass_x', 'Survived_x']],
          hue='Survived_x'
        )

plt.show()
```



```
In [71]: sns.boxplot(
    x='Pclass_x',
    y='Fare_x',
    hue='Survived_x',
    data=df
)

plt.show()
```

